

# The quantum Hall effect

Landau levels, quantized transport, and fractional charge

11 September 2026

The quantum Hall effect connects a measurable transport coefficient to the structure of a many-electron state. The integer effect can arise without electron interactions; fractional states require them. This note follows the common starting point and then separates the two mechanisms.

## 1 Landau levels and their degeneracy

Consider spin-polarized electrons of charge  $-e$ , with  $e > 0$ , effective mass  $m_*$ , and a uniform perpendicular field of magnitude  $B > 0$ . For a parabolic band,

$$H = \frac{(\mathbf{p} + e\mathbf{A})^2}{2m_*}, \quad E_n = \hbar\omega_c \left(n + \frac{1}{2}\right), \quad \omega_c = \frac{eB}{m_*}, \quad n = 0, 1, \dots \quad (1)$$

Here  $\nabla \times \mathbf{A}$  is the magnetic field; reversing its direction reverses the Hall response but not these energies. The magnetic length and flux quantum are

$$\ell_B = \sqrt{\frac{\hbar}{eB}}, \quad \Phi_0 = \frac{h}{e}. \quad (2)$$

For area  $A$ , a Landau level contains  $N_\Phi = BA/\Phi_0 = A/(2\pi\ell_B^2)$  orbitals, up to boundary corrections. One way to count them is to use Landau gauge: allowed momenta are spaced by  $2\pi/L_y$ , while the corresponding guiding centers are spaced by  $2\pi\ell_B^2/L_y$  across a sample of width  $L_x$ .

With  $N$  electrons and density  $n_e = N/A$ , define the filling factor

$$\nu = \frac{N}{N_\Phi} = 2\pi\ell_B^2 n_e. \quad (3)$$

Integer filling fills an integer number of levels in the clean, noninteracting, spin-resolved problem. Zeeman and valley degeneracies must be treated separately when applying this count to a material.

## 2 What makes a plateau robust?

For  $p$  filled levels, the ideal low-temperature response is  $|\sigma_{xy}| = pe^2/h$  and  $\sigma_{xx} = 0$ . The sign depends on field and tensor conventions. More generally, for a periodic noninteracting two-dimensional insulator,

$$\sigma_{xy} = \frac{e^2}{h} C, \quad C = \frac{1}{2\pi} \sum_{n \in \text{occ}} \int_{\text{BZ}} \Omega_n d^2k \in \mathbb{Z}, \quad (4)$$

where  $\Omega_n$  is the Berry curvature with the matching orientation convention. This is the band-topological expression for the integer response.

Disorder can localize bulk states between extended states, allowing density to change over a range without changing the Hall plateau. Thus a mobility gap, rather than a perfectly empty spectral gap, can suffice. The clean Bloch-band formula is not itself a derivation of localization or of a finite plateau width.

### 3 Why a fractional plateau is a many-body problem

At a partially filled, dispersionless Landau level, kinetic energy no longer selects a unique state. Interactions act within the macroscopically degenerate subspace. A model wavefunction for an incompressible state at  $\nu = 1/m$ , with  $m$  a positive odd integer, is the Laughlin state [2]:

$$\Psi_m(z_1, \dots, z_N) = \mathcal{N} \prod_{i < j} (z_i - z_j)^m \exp \left[ - \sum_i \frac{|z_i|^2}{4\ell_B^2} \right]. \quad (5)$$

Here  $z_i = x_i + iy_i$ ,  $\mathcal{N}$  normalizes the state, and the field orientation is chosen so the lowest-level orbitals are holomorphic. The integer  $m$  is unrelated to the effective mass  $m_*$ . Odd  $m$  makes the state antisymmetric under exchange; the repeated zero suppresses close particle encounters. Counting the largest polynomial degree gives  $N_\Phi \simeq mN$ , hence  $\nu \rightarrow 1/m$  at large  $N$ .

A quasihole at  $\eta$  is represented by

$$\Psi_{\text{qh}} = \prod_i (z_i - \eta) \Psi_m. \quad (6)$$

For a gapped, screened Laughlin liquid, its integrated electron deficit corresponds to charge  $+e/m$ . This is a collective excitation of the correlated state. It does not split an individual electron into isolated pieces.

### 4 Flux insertion and an effective response

On an annulus, adiabatically insert one flux quantum through the hole. Faraday's law produces an azimuthal electric field; the Hall current transfers charge between the boundaries. Integrating the radial current gives, in magnitude,

$$|\Delta Q| = |\sigma_{xy}| \Phi_0 = \nu e. \quad (7)$$

This gives one electron per filled integer level, or  $e/m$  in a Laughlin state. The argument assumes an appropriate adiabatic evolution that keeps bulk excitations suppressed.

The same low-energy response can be encoded by a compact emergent gauge field  $a$ . In units  $e = \hbar = 1$ , with external electromagnetic connection  $A$ , choose

$$S[a, A] = -\frac{m}{4\pi} \int a \wedge da + \frac{1}{2\pi} \int A \wedge da. \quad (8)$$

Locally, the  $a$  equation gives  $m da = dA$ . Substituting the solution yields  $S_{\text{eff}}[A] = (4\pi m)^{-1} \int A \wedge dA$ , which reproduces  $\sigma_{xy} = e^2/(mh)$  in this orientation. This local response calculation leaves global sectors and boundary degrees of freedom to be handled separately.

**Distinction to remember.** An integer Hall state is characterized by a quantized band response; a fractional Hall state additionally has intrinsic topological order, including fractional excitations and topology-dependent ground-state degeneracy. A measured plateau is a starting point for identifying the state, not a complete specification of all its quasiparticles.

## References

- [1] D. Tong, *Lectures on the Quantum Hall Effect* (2016). [arXiv:1606.06687](https://arxiv.org/abs/1606.06687).
- [2] R. B. Laughlin, "Anomalous Quantum Hall Effect: An Incompressible Quantum Fluid with Fractionally Charged Excitations," *Phys. Rev. Lett.* **50**, 1395 (1983). [doi:10.1103/PhysRevLett.50.1395](https://doi.org/10.1103/PhysRevLett.50.1395).